

EverCare System Documentation

A Comprehensive Technical Guide, Architecture & Database Specification

Project: EverCare Flutter Mobile App	Target Audience: Seniors & Caregivers
Framework: Flutter 3.11+ / Dart	Backend: Supabase (Auth, DB, RLS, Edge)
SDG Alignment: UN SDG 3 Good Health	Author: Brent Bernardo

1. Executive Summary & Objective

EverCare is a Flutter care companion tailored for Filipino seniors, family members, and caregivers. It consolidates vital health metrics—such as blood pressure readings, medicine intake schedules, medical appointments, personal health journals, emergency contact data, and trusted care relationships—into a single, accessible, and elderly-friendly mobile application.

The application actively supports **United Nations Sustainable Development Goal 3: Good Health and Well-Being** by lowering barriers to recording, reviewing, and coordinating everyday health information. EverCare functions strictly as a supportive care tool; saved blood pressure readings are clearly flagged as non-medically verified to avoid misinterpretation.

***Medical & Emergency Disclaimer:** EverCare is not a diagnostic tool, clinical device, or emergency response system. All measurements, range classifications, and AI insights are provided for educational and organization purposes only.*

2. System Architecture

EverCare employs a multi-tiered, decoupled architecture comprising a Flutter mobile client, owner-scoped Supabase backend, Bluetooth Low Energy (BLE) hardware integrations, OpenStreetMap services, and serverless AI edge functions.

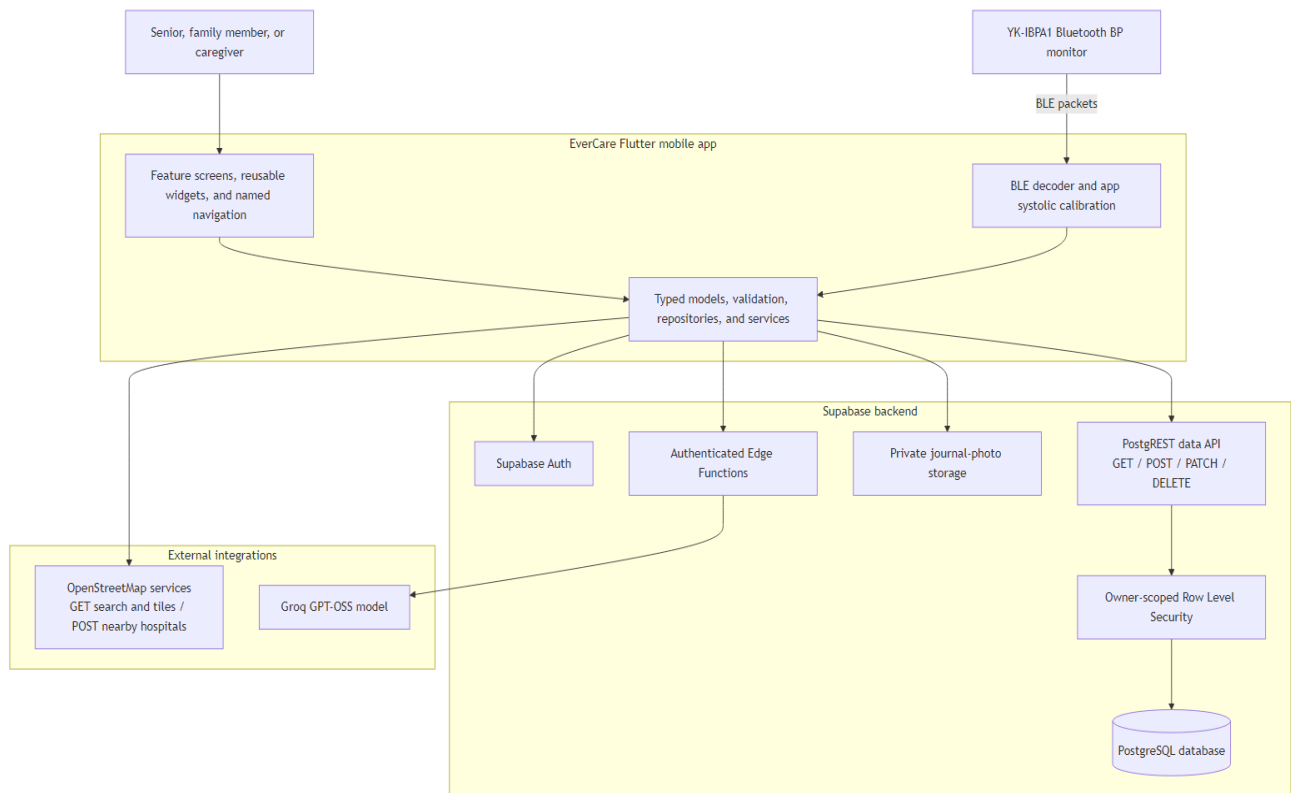


Figure 2.1: EverCare System Architecture & Component Interaction

- **Flutter Mobile App:** Houses 15 screen modules, typed Dart domain models, repositories, and an elderly-friendly Material theme.
- **YK-IBPA1 BLE Monitor:** Connects directly via Bluetooth Low Energy to stream hardware packets and decode blood pressure values.
- **Supabase Backend:** Manages Auth (email/password), PostgREST RESTful APIs, owner-scoped Row Level Security (RLS), and private journal media storage.
- **Edge Functions (Groq LLM):** Executes authenticated serverless requests to generate contextual blood pressure summaries without exposing API credentials to the client.
- **OpenStreetMap Infrastructure:** Integrates Overpass API (nearby hospitals), Photon (debounced search autocomplete), and Nominatim (geocoding).

3. Database Design & Security Specifications

The EverCare database is deployed on PostgreSQL via Supabase. It strictly enforces data privacy using Row Level Security (RLS) policies on all 10 personal tables, ensuring authenticated users can only interact with their own owned records or explicitly granted care connections.

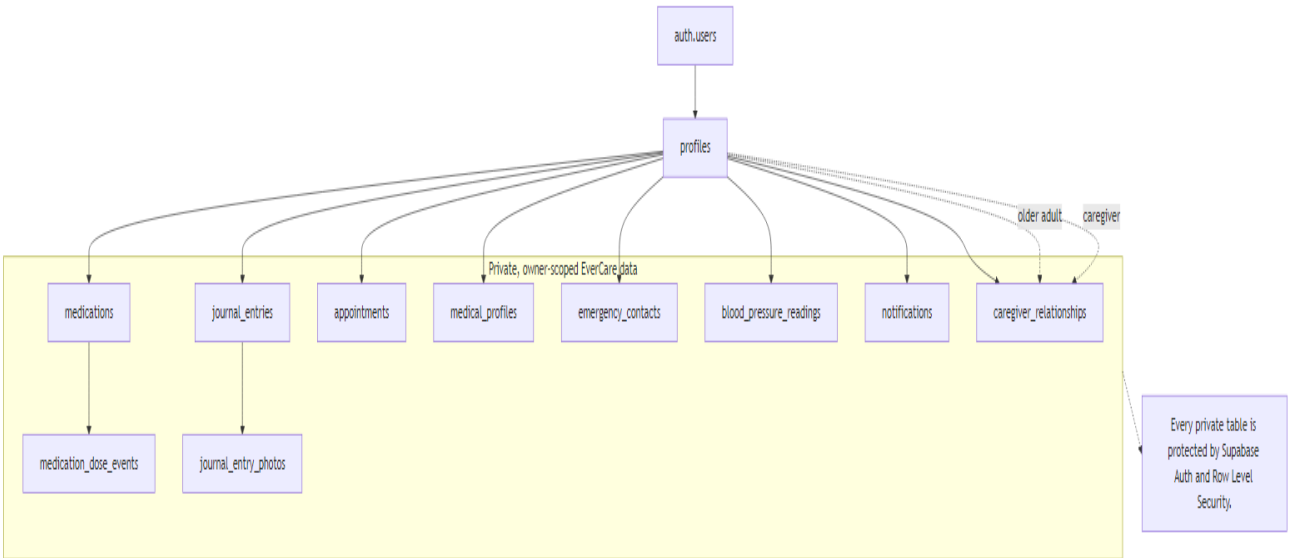


Figure 3.1: Database High-Level Schema Overview

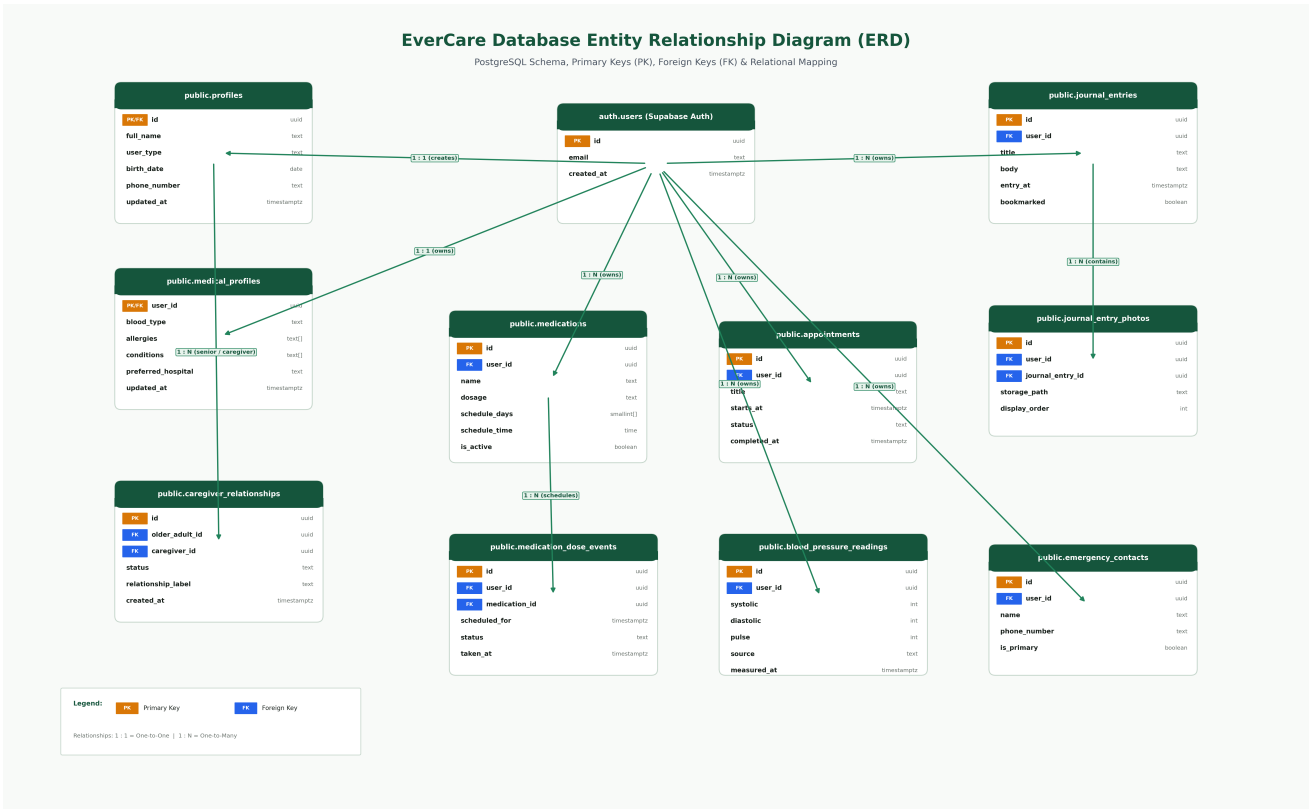


Figure 3.2: Entity Relationship Diagram (ERD) & Foreign Keys

3.1 Core Database Tables Summary

Table Name	Primary Key / FK	Description & Security Constraints
profiles	id (PK -> auth.users)	User full name, user_type ('senior','caregiver','family_member'), birth date, and phone number. Auto-created via trigger on user signup.
medical_profiles	user_id (PK -> auth.users)	Stores blood type, allergies[], medical conditions[], and preferred hospital. Owner-scoped RLS.
medications	id (PK), user_id (FK)	Medication details, schedule_days (ISO 1-7), schedule_time, dosage, and active status flag.
medication_dose_events	id (PK), medication_id (FK)	Dose adherence log. Tracks status ('scheduled', 'taken', 'skipped', 'missed') and taken_at timestamp.
appointments	id (PK), user_id (FK)	Doctor visits, starts_at, clinic, address, and status ('upcoming','completed','missed','cancelled'). 24h missed auto-reconciliation.
journal_entries	id (PK), user_id (FK)	Health journal entry with title, body, mood, symptoms[], activities[], and bookmark state.
journal_entry_photos	id (PK), journal_entry_id (FK)	Storage paths for attached photos uploaded to private Supabase storage bucket.
blood_pressure_readings	id (PK), user_id (FK)	Systolic, diastolic, pulse, measured_at, source ('ble','manual'), raw packet hex. Immutability enforced via trigger.
emergency_contacts	id (PK), user_id (FK)	Emergency contact details and relationship. Unique partial index enforces one primary contact per user.
caregiver_relationships	id (PK), older_adult_id, caregiver_id	Links senior to caregiver with status ('pending','accepted','declined','revoked'). State transition guards enforced.
notifications	id (PK), user_id (FK)	In-app notifications for overdue medicines, appointments, and system alerts.

4. Core Features & Business Logic Implementation

4.1 BLE Blood Pressure Hardware Capture

- Integrates real YK-IBPA1 Bluetooth Low Energy blood pressure monitors using flutter_blue_plus.
- Parses hardware packet notifications, extracts systolic, diastolic, and pulse parameters via YkIbpa1PacketDecoder.
- Applies a calibrated app-level systolic offset adjustment before displaying readings to the user.
- Requires explicit user action ('Save Reading') before writing to Supabase; unconfirmed readings remain transient.
- Immutability DB Trigger blocks updates to measurement capture fields once stored.

4.2 Medication Adherence Tracking

- Supports recurring weekday schedules (Monday through Sunday ISO array selection).

- In-app adherence reconciler automatically marks scheduled doses as 'Missed' when overdue by over 1 hour while the app is active.
- Enforces server-clock validation via PostgreSQL functions to prevent forged device timestamps.
- Tracks completed medication courses, archiving finished prescriptions while retaining complete historical dose logs.

4.3 Appointment Management & Attendance Reconciliation

- Schedules medical appointments with visit times, doctor names, specialties, and clinic locations.
- Appointments transition to 'Missed' exactly 24 hours after visit time if left uncompleted.
- Caregivers/seniors can execute 'Mark Done Late' to reconcile missed visits with actual attendance notes.
- Integrates external Google Maps directions link for clinic navigation.

4.4 OpenStreetMap Hospital Finder

- Leverages flutter_map for interactive, keyless map rendering using OpenStreetMap tiles.
- Queries nearby hospitals via Overpass API with multi-endpoint failover and rate limiting.
- Provides search-as-you-type hospital autocomplete using Photon (debounced at 3 characters).
- Uses Nominatim for full geocoding of explicitly submitted search terms.

4.5 AI Health Insights & Edge Functions

- Invokes authenticated Supabase Edge Functions connected to Groq LLM API.
- Transmits non-identifying health parameters (BP numbers and range category) to generate simplified summaries.
- Keeps API credentials strictly inside serverless environment secrets.

5. Project Architecture & Codebase Structure

EverCare follows clean architecture principles with clear separation between UI, state management, business logic repositories, and remote data services.

Directory / Component	Responsibilities
lib/config/	Public runtime configurations (Supabase URL & Publishable Key).
lib/decoders/	Low-level YK-IBPA1 BLE binary packet decoder.
lib/models/	18 typed Dart domain classes (BloodPressureReading, Medication, etc.).
lib/repositories/	9 owner-scoped data access repositories handling Supabase PostgREST CRUD.
lib/routes/	Centralized named navigation route generator (AppRoutes) and route observers.
lib/screens/	15 feature pages (Home dashboard, Health, Medications, Appointments, Journals, Emergency, etc.).
lib/services/	11 service controllers: Auth, BLE monitor, Hospital Finder, AI, Accessibility.
lib/theme/	Design system tokens: AppColors, AppTextStyles, AppMotion, AppTheme.
lib/widgets/	17 reusable UI components (cards, skeleton loaders, glass surfaces, headers).
supabase/migrations/	5 versioned PostgreSQL migration files establishing schema, triggers, and RLS policies.
assets/	Local healthcare graphics, brand logos, BP category visuals, and NIA Caregiver Handbook.

6. Quality Assurance, Testing & Verification Results

EverCare undergoes rigorous static analysis and automated verification to ensure codebase reliability, API correctness, and database policy compliance.

- ✓ **Static Code Analysis:** Validated via flutter analyze enforcing flutter_lints 6.0.0 with zero remaining warnings or errors.
- ✓ **Automated UI & Logic Smoke Tests:** Executed via flutter test covering UI navigation, model transformations, repository serialization, and BLE service state machines.
- ✓ **CRUD API Verification:** Validated with automated PowerShell verification scripts testing full Create, Read, Update, Delete lifecycles over PostgREST endpoints.
- ✓ **RLS Security Audit:** Verified zero cross-tenant data leakage by testing queries across distinct authenticated user accounts.

EverCare System Readiness Summary: The system successfully passes all Week 6 midterm requirements. The application delivers a secure, accessible, and robust care companion experience for Filipino seniors and their care networks.